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A multiagent-based modeling approach for emergency evacuation plan optimization during toxic gas releases within chemical plants

Chaoying Li [1], Jie Hou [1], Wuyi Cheng [*], Wenmei Gai [*]

School of Engineering and Technology, China University of Geosciences (Beijing), Beijing 100083, China

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ABSTRACT

It is essential to minimize the gas exposure risk during evacuation in the event of toxic gas releases. This study uses a multiagent-based model to simulate the evacuation processes of a chemical plant and surrounding residential areas. An evaluation and optimization method for evacuation plans was proposed. A risk assessment model for ambient evacuation was established, and the evacuation plan optimization process was proposed. Using a chemical plant in Tianjin, China, as an example, the evacuation scenario of workers and surrounding residents after liquid ammonia release was simulated, and the effectiveness of the optimization strategies for the original evacuation plan is verified. The results reveal that using optimization strategies dramatically reduces the gas exposure risk during evacuation. The gas exposure risk during evacuation of some evacuation sub-areas is reduced by optimizing the evacuation route. Meanwhile, issuing the evacuation order sooner and optimizing the warning diffusion model exerts an excellent risk reduction effect on areas with low ammonia concentrations. Evacuation optimization strategies suitable for different evacuation sub-areas are not the same. They need to be determined according to the real situations of the evacuation sub-areas. The results can help facilitate decision-making for the formulation and optimization of emergency plans for toxic gas releases.

1. Introduction

In the production, operation, storage, transportation, use, and abandonment of hazardous chemicals, toxic gas release accidents refer to the unintentional release of toxic gas due to human factors, equipment factors, management, or environmental factors posing a threat to human health and environmental stability (Bo and Chen, 2010; Haastrup and Rasmussen, 1994). After the accident, the toxic gas will spread into the environment in a toxic gas cloud. In extreme cases, toxic gas clouds may also spread to regions outside the chemical industry area, such as the surrounding residential or commercial areas. If personnel in the surrounding facilities of the industrial plants fail to take evasive action, they will be directly exposed to the toxic gas cloud causing casualties (Li et al., 2009a, 2009b; He et al., 2011; Georgiadou et al., 2010; Dou et al., 2019).

In the emergency response for tackling toxic gas release accidents, the most common protective actions for the population are sheltering-in-place (remaining in one's structure or location) and evacuation (Chakrabarti and Jigisha, 2011, 2013; Georgiadou et al., 2010). It is true that special shelters have an excellent shielding effect on toxic gas and effectively protect people's safety (Jetter and Whitfield, 2005). However, the shielding effect of toxic gases is limited in ordinary residential buildings and even lower in industrial facilities (Argyropoulos et al., 2017). Evacuation is the most effective protective action to ensure the safety of people when shelter-in-place cannot provide adequate protection for the threatened people. An "emergency evacuation planning area" (EEPA) is usually determined for such accidents in China. The scope of the EEPA, which is included in the emergency planning zone, refers to an area where an order needs to be given to evacuate the people to a specific location after an accident. People within the area are usually advised to evacuate from the risky area to the safe area in an emergency. During the evacuation, it is particularly important to guide people in order to evacuate the threat area quickly and safely after the accident (Li et al., 2011).

Performing emergency drills and formulating emergency plans regularly before the accident and implementing the emergency plan when the accident occurs, is an effective measure to improve the emergency response capabilities of people (Tseng et al., 2008).

* Corresponding authors.
E-mail addresses: cwy@cugb.edu.cn (W. Cheng), gaiwenmei@cugb.edu.cn (W. Gai).
[1] These authors contributed equally to this work and should be considered co-first authors.

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However, in the chemical industry sector, an EEPA may involve tens of thousands of people in multiple buildings and communities (Dou et al., 2019; Georgiadou et al., 2010; Wang et al., 2012). Such a large-scale emergency evacuation drill is difficult to implement (Li et al., 2009; Gai et al., 2018; Gai and Deng, 2018). The households' participation in evacuation drills during the planning phase differs significantly from evacuations during emergencies. Households rarely participate in evacuation drills that emergency response organizations hold during the planning period. However, in the case of an evacuation during real emergencies, the households must be involved. Therefore, the participants of the evacuation drills are different from those of real emergency evacuations (Lindell and Perry, 1980, 2006; Lindell, 1995). Conversely, evacuation behavior analysis based on evacuation cases in real toxic gas release accidents is useful for formulating emergency response plans (Hou et al., 2020a, 2020b). Moreover, it is difficult to cover all the possible emergency scenarios according to the backtracking and extrapolation of real evacuation scenarios. Evacuation simulations based on evacuation modeling technology can overcome the above limitations and can be used to analyze the evacuation behaviors under different gas release scenarios; thus, helping to predict possible problems within an emergency evacuation, optimize measures in advance, and improve evacuation efficiency.

Early studies of evacuation models focused on abstract road network nodes rather than individual behavior (Lindell, 2008; Borrmann et al., 2012). However, the existing research shows that people's behavior after receiving the evacuation order results from their risk perception (Mccaffrey et al., 2018; Mileti and Peek, 2000). The people's perception of risk is affected by many factors, such as receiving the evacuation advisories from authorities (Huang et al., 2016) or perceiving the environmental cues of the accident (explosion sound, pungent smell, etc.) (Tierney, 2009; Huang et al., 2012; Mccaffrey et al., 2018). Apart from relying on the obtained information about the accident, people base their perception of the risks on their previous experience of similar incidents and their neighbors' reactions; moreover, such perception changes over time (Lindell, 2008; Lindell et al., 2005). Therefore, evacuation is a complex process involving individual decision-making and mutual influences (Hou et al., 2020b; Lindell and Perry, 2004; Mccaffrey et al., 2018).

In the process of evacuation modeling, a multiagent-based evacuation simulation technology can configure diverse individual characteristics in order to mimic people's interaction responses and the environment's influencing mechanism upon the individuals. Simulation technology is commonly used to design and optimize emergency evacuation strategies for industrial accidents due to its low staffing and material expenses (Wagner and Agrawal, 2014; Poulos et al., 2018; Dou et al., 2019).

Gas exposure risk during evacuation (We will use the shorter expression "evacuation exposure risk" throughout the rest of the article) assessment is another critical component of emergency evacuation preparations (Chakrabarti and Jigisha, 2013). Before issuing evacuation orders, the decision-makers should examine the dangers and potential threats in specific locations to maximize emergency resources and improve emergency evacuation plans (Lindell, 1995). Concentration discrimination (Cui et al., 2008) is the current method for determining evacuation exposure risks in toxic gas releases, but it has limitations. The premise of this method is that an evacuation path should be directed away from the release source. However, in some actual situations, evacuation paths direct people toward the risk area exit (Lindell and Prater, 2007) closer to the release source first before finally evacuating them to a safe area. As shown in Fig. 1, people A in a closed ESR can only be evacuated through the sole risk area exit. In this circumstance, evacuees' peak exposure would be higher but their exposure duration could be lower than if they remained at their original location for the duration of the release. In the evacuation process, individuals are in motion, and their locations at different times and the corresponding exposure concentrations tend to vary. Hence, failure to consider these factors may lead to an evaluation outcome that is inconsistent with reality. If a multiagent-based modeling technology is combined with the concentration discrimination method, the individual characteristics can be integrated into the evacuation exposure risk assessment. In this way, the limitations of the current evacuation exposure risk assessment method can be avoided.

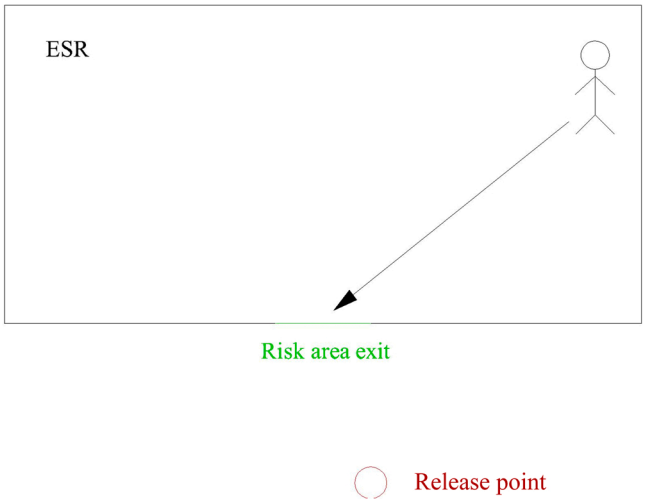


Fig. 1. Schematic diagram of residents' escape in a certain situation.

In this paper, we simulate an evacuation procedure in a chemical plant and surrounding residential areas using multiagent modeling. A quantitative evacuation exposure risk calculation approach is proposed to estimate the evacuation exposure risk of each location under the original evacuation plan in combination with the process of toxic gas diffusion and individual evacuation, as well as considering the individual evacuation characteristics. The ALARP (As Low as Reasonably Practical) principle is introduced to assess the evacuation exposure risk, and re-plan the evacuation plan for places where the evacuation exposure risk estimate findings do not fulfill the preset risk control thresholds. The optimization of the evacuation plan tested three strategies: improving the evacuation route, evacuation order issuance time, and the warning diffusion model. A single or a combined strategy was determined in order to optimize the original evacuation plan that needs to be re-planned and try to make the optimization results meet the preset risk control thresholds. It should be pointed out that the chemical plant selected in this paper mainly exists as a release source to simulate the emergency response of the workers in the plant and the surrounding residents when a release accident occurs. Therefore, when designing the accident scenario, this paper refers to a chemical plant in Tianjin, China, and mainly emphasizes its geographical location, surrounding environment and population characteristics, while the actual production process and structure in the chemical plant are not the focus of this paper. The results are expected to facilitate decision-making regarding the formulation and optimization of emergency plans for toxic gas releases.

The rest of this paper is structured as follows. *Section 2* models the evacuation process based on a multiagent model. *Section 3* establishes the risk assessment model of an emergency evacuation plan. The optimization process of the evacuation plan is proposed based on the evaluation of evacuation route selection, evacuation order issuance time, and warning diffusion model. *Section 4* presents the results of a case study, and finally, *Section 5* concludes the paper.

2. Multiagent-based evacuation processes modeling

Stepanov and Smith (Stepanov and Smith, 2009) presented a clear division of the stages of evacuation time. This division is most recently presented in (Lindell et al., 2019a, 2019b) and applied to hurricane

evacuations in (Lindell et al., 2020, 2021; Sorensen et al., 2020) , providing a clearer division of the elements within this process. However, the evacuation scenarios of the toxic gas release accidents outlined in this paper have certain particularities. The toxicity and dispersion properties of the toxic gas after the release requires us to pay more attention to the details of the evacuation process. Therefore, in order to refine the description of individual evacuation process, the evacuation process modeling in this paper consists of two sub-processes: pre-evacuation process modeling and evacuation movement process modeling. The pre-evacuation process starts from when the accident occurs until the individual takes their evacuation actions (Fig. 2).

2.1. Modeling of pre-evacuation process

The pre-evacuation process involves not only the individual decision-making but also the interaction between individuals. As it represents a complicated and large-scale adaptive system, a complex network model is adopted herein (Fig. 3). This model is a mathematical generalization of the distinction between the "broadcast process" and "contagion process", which was originally introduced by Rogers and Sorensen (Rogers and Sorensen, 1988), and then continuously developed (Stone et al., 2007; Wang et al., 2012, 2018; Zhang and Tian, 2008). First, a digraph emergency communication network $G(V, A, W_v, W_a)$ is constructed based on the complex network models (such as the Erdos–Renyi (ER) random network and the small world (WS) network). Here, process I primarily shows the broadcast process, and process II primarily shows the contagion process. $V = (v_i | i = 1, 2, \ldots, n)$ is the set of nodes. v_0, $v_i(i = 1, 2, \ldots, s)$, and $v_i(i = s+1, s+2, \ldots, n)$ are defined as the super source, source, and ordinary nodes, and they correspondingly denote the decision-makers that release the evacuation order, the personnel involved in emergency operations, and the ordinary individuals in the emergency response process. $A \sqsubseteq V \times V$ is the set of arcs. $W_v = \{e_i(t), c_i, l_{di}, l_{evi}\}$ denotes the set of weights of node v_i, where $e_i(t)$ denotes the information value of v_i at the current moment t, $e_i(t) \in [0, 1]$. $e_0(t) = 1$ indicates that the decision makers have issued the protective action recommendations that advise people to evacuate and provide evacuation traffic management to facilitate traffic flows. $e_0(t) = 0$ indicates that the decision makers haven't yet issued protective action recommendations. $c_i \in [0, 1]$ is the possibility that node v_i has herd behavior. $l_{di}, l_{evi} \in [0, 1]$ are threshold parameters set for $v_i(i = 1, 2, \ldots, n)$, and they correspondingly denote when the value of $e_i(t)$ exceeds one of the above two thresholds, node v_i will send evacuation information to neighboring nodes (l_{di}) or decide to evacuate by himself (l_{evi}). $W_a = \{b_{ij}\}$ denotes the set of weights of arc (v_i, v_j). $b_{ij} \in [0, 1]$ denotes how much node v_i trusts node v_j, which is related to the emergency communication channels and the closeness of the relationship between them.

After the decision makers issue the evacuation order, v_0 first delivers the evacuation order to the personnel involved in the emergency operations, namely $v_i(i = 1, 2, \ldots, s)$, and then the dissemination of evacuation warnings (order/messages) starts. During this process, people's attitude towards the information they obtained is constantly developing with an increase in the amount of information obtained. When the value of accident information obtained increases to a certain level, it will prompt people to make decisions such as transmitting evacuation warnings to surrounding people and starting the evacuation movement. In the model herein, the diffusion of the evacuation warnings obeys the hypotheses described as follows (Stone et al., 2007).

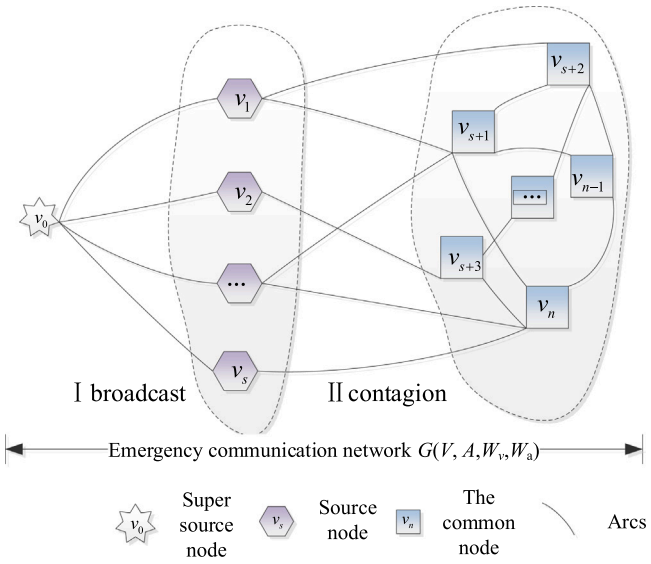


Fig. 3. Structure of emergency communication network $G(V, A, W_v, W_a)$.

i. If the value of information obtained by node $v_i(i = 0, 1, 2, \ldots, n)$ exceeds its dissemination threshold l_{di}, node v_i will transmit evacuation warnings to the adjacent node v_j. Thus, the information value of node v_j can be estimated according to Eqs. (1) and (2).

$$e_j(t_{m+1}) = e_j(t_m) + \big(e_i(t_m) - e_j(t_m)\big) \cdot a_{ij}(t),\ e_i(t_m) > l_{di} \tag{1}$$

$$a_{ij}(t) = b_{ij} c_i \big(1 - e_j(t)\big) \tag{2}$$

where $a_{ij}(t)$ denotes the probability of node v_i transmitting evacuation warnings to node v_j.

ii. If the information value obtained by node $v_i(i = 0, 1, 2, \ldots, n)$ exceeds the evacuation decision-making threshold, node v_i will make an evacuation decision, which indicates the completion of the pre-evacuation process.

We can assume that the time when the accident occurs is Zero point t_0, and the evacuation order is issued at t_1. According to the simulation based on the above established pre-evacuation process model, the curve of the evacuation warning diffusion of a specific evacuation sub-area

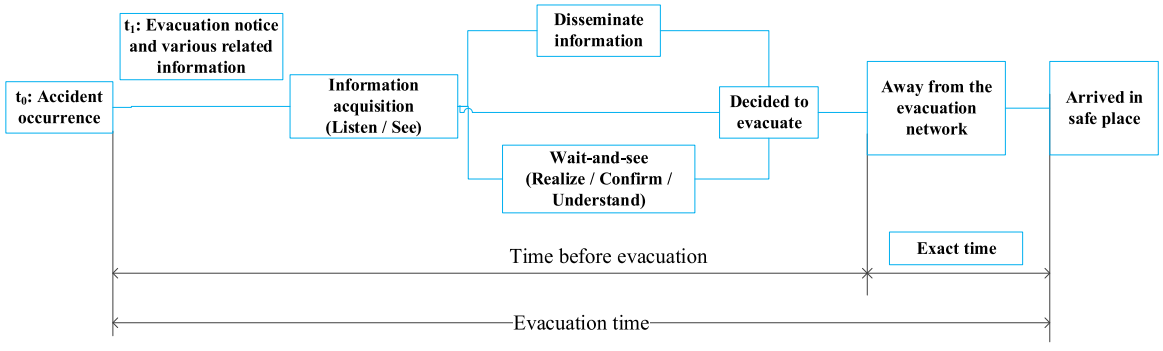


Fig. 2. Individual evacuation process.

$x(x = 1, 2, \ldots, N_{sub})$ can be obtained, as described in Eq. (3). Here, $x_i(t)$ denotes whether node v_i decides to evacuate at the moment of t. $x_i(t) = 1$ denotes node v_i decides to evacuate at the moment of t; $x_i(t) = 0$ denotes node v_i decides not to evacuate at the moment of t.

$$f_{\text{pre}}(t) = \begin{cases} 0, & t \le t_0 \\ \dfrac{\sum_{i=1}^{n} x_i(t)}{n}, & t > t_0 \end{cases} \tag{3}$$

$$x_i(t) = \begin{cases} 1, & e_i(t) > l_{evi} \\ 0, & e_i(t) \le l_{evi} \end{cases} \tag{4}$$

2.2. Modeling of evacuation movement process

Pathfinder is an intelligent emergency evacuation simulator developed by Thunderhead Engineering Company in the United States. It can simulate individual evacuation movements under various scenarios. Here, we use Pathfinder and the analysis results of the pre-evacuation process to build the evacuation movement process model (Fig. 4).

The EEPA is divided into N_{sub} relatively independent evacuation subregions (ESR) with smaller areas (Gai et al., 2018). Thus, the evacuation environment can be established based on the EEPA division on the GIS map and the building parameters. Then, according to the predictions of the population size to be evacuated in the ESRs, the agents representing the individuals are placed inside and outside the buildings in the ESRs following a certain proportion. According to the demographic survey data, the basic attribute parameters, including gender, age, and speed, are set for the individuals in the ESRs (Hu, 2018). The individual evacuation delay process is set according to Eqs. (3) and (4). Finally, the individual evacuation movement mode in the EEPA and ESRs is determined in order to simulate the evacuation movement process.

3. Evaluation and optimization of evacuation plans based on evacuation exposure risk assessment

3.1. Definition of individual evacuation scenarios

Given different risk perception levels and environmental conditions, the time when individuals take evacuation actions may vary. Moreover, because of differences in preferences and the influence of environmental factors during an evacuation, the evacuation population may choose different evacuation directions and complete the evacuation along multiple evacuation paths (Georgiadou et al., 2007; Stepanov and Smith, 2009). Therefore, there may be different scenarios in the evacuation process. For the convenience of calculation, the evacuation time is divided into multiple time intervals, and the people who begin to evacuate from a specific ESR within a time interval are categorized into a group. Evacuees from each group are further divided into even smaller groups according to the evacuation routes they choose. We define the evacuation of such a small group as an individual evacuation scenario.

3.2. Probability estimation of individual evacuation scenarios

Through the simulation of ambient evacuation, the loading rate curve of evacuees $f_{\text{dpj}}(t)$ in a specific risk area exit$j(j = 1, 2, \ldots, N_{exi})$ of any ESR can be obtained. $P_{\text{ev}} = \{path_{j,l} | \ l = 1, 2, \ldots, N_{path,j}\}$ denotes the set of evacuation paths chosen by the evacuees leaving from the exit j. NP_j and $NP_{j,l}$ denotes the number of evacuees leaving from the exit j and the path $path_{j,l}$ correspondingly. According to the definition of individual evacuation scenarios in *Section 3.1* , the probability p_k related to a specific evacuation scenario k can be estimated as follows:

$$p_k = \begin{cases} p_{\text{dpj}}(t_m) \cdot \dfrac{NP_{j,l}}{NP_j}, & k = 1, 2, ., N_s - 1 \\ 1 - \sum_{j=1}^{N_{\text{exi}}} f_{\text{dpj}}\left(t_{N_{\text{prej}}}\right), & k = N_s \end{cases} \tag{5}$$

$$p_{\text{dpj}}(t_m) = f_{\text{dpj}}(t_m) - f_{\text{dpj}}(t_m - \Delta t) \tag{6}$$

$$N_s = \sum_{j=1}^{N_{\text{exi}}} N_{\text{prej}} \cdot N_{\text{path},j} + 1 \tag{7}$$

where $p_{\text{dpj}}(t_m)$ is the probability that an individual of the ESR under study leaves from the exit j within the interval $[t_m - \Delta t, t_m]$. Δt denotes a specific time interval, $m = 1, 2, \ldots, N_{prej}$. N_{prej} denotes the number of time intervals divided. N_s is the number of individual evacuation scenarios of the ESR under study.

3.3. Calculation of health consequences in evacuation

Health consequences during the evacuation related to a specific in-

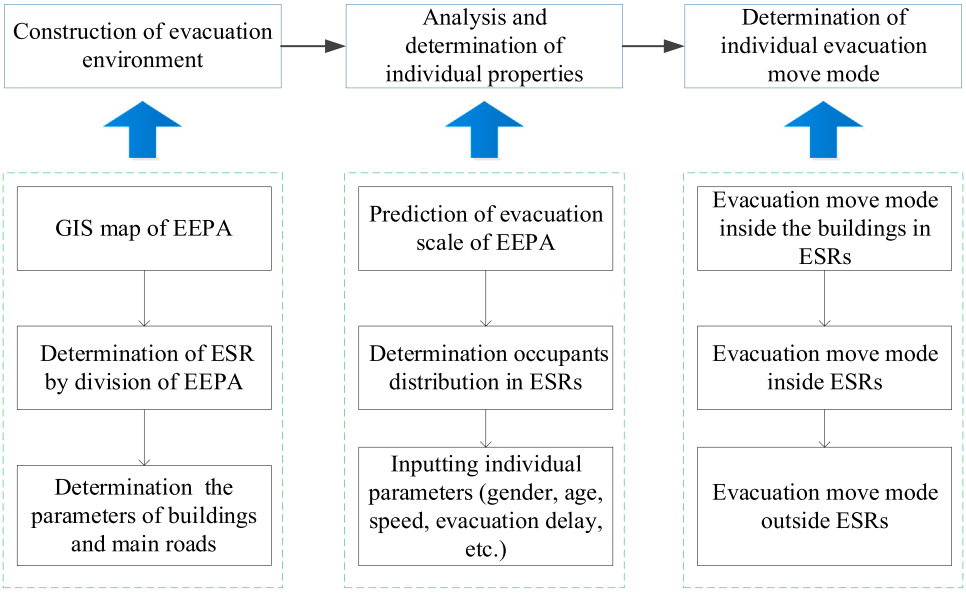


Fig. 4. Modeling framework of the evacuation movement process based on the Pathfinder software.

dividual evacuation scenario $k(k = 1, 2 \ldots N_s)$ of an ESR x $(x = 1, 2 \ldots N_{sub})$ can be measured by the individual death probability $d_{x,k}$. $d_{x,k}$ is estimated through the "probit function" (Papazoglou et al., 1992) as follows:

$$d_{x,k} = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{Y_k - 5} \exp\left(-\frac{u^2}{2}\right) \mathrm{d}u \tag{8}$$

$$Y_k = \beta_0 + \beta_1 \cdot \ln V_{x,k} \tag{9}$$

$$V_{x,k} = \int_0^{t_m} {c_{x,t}}^n \mathrm{d}t + \int_{path_{j,l}} {c_{(y,z),t}}^n \mathrm{d}t \tag{10}$$

$$t_m = f(w, t_r) \tag{11}$$

Where u is an integral variable; $V_{x,k}$ is the probability variable; β_0, β_1, and n are constants related to the type of chemicals; $V_{x,k}$ is the lethal dose related to the individual evacuation scenario. k; $c_{x,t}$ is the average concentration of toxic gas in ESR x at time t; $c_{(y,z),t}$ is the concentration of toxic gas at coordinated (y, z) of the evacuation path $path_{j,l}$ at time t; t_m denotes the moment when people to be evacuated leave ESR x. The value of t_m depends on the warning diffusion model W_x of the evacuation warnings in ESR x and the issuance time t_r of the evacuation order.

3.4. Calculation of evacuation exposure risk

Suppose $P = \{W, T, R\}$ denotes an evacuation plan. Such a plan consists of the warning diffusion model $W = \{W_x, x = 1, 2, \ldots, n\}$, the release time of the evacuation order $T = \{t_{m,x}, x = 1, 2, \ldots, n\}$, and the evacuation path assignment scheme $R = \{P_{ev,x}, x = 1, 2, \ldots, n\}$. The evacuation exposure risk $ER(P)$ under a specific evacuation plan $P = \{W, T, R\}$ can be calculated through Eqs. (12) and (13).

$$ER(P) = \sum_{x=1,P}^{n_s} \rho_x \cdot ER_x(p) \tag{12}$$

$$ER_x(P) = \sum_{k=1,P}^{N_s} p_k \cdot d_k \tag{13}$$

where $ER_x(P)$ is the evacuation exposure risk of ESR x under the evacuation plan P. ρ_x is the proportion of people in ESR x to that of people in the EEPA.

3.5. Acceptable standards for evacuation plans

It is impractical to optimize an evacuation plan by reducing the evacuation exposure risk without considering the cost. Thus, the concept of acceptable risk is often used (Georgiadou et al., 2010; Jonkman et al., 2003; Xu and Fan, 2014). Fischhoff (Fischhoff, 1983) pointed out that risk is not accepted unconditionally and could only be accepted when the benefits obtained can compensate for the risks. In the final analysis, the acceptable risk is a decision problem. The decision is what creates acceptable risk, not the risk itself. Britain's Health and Safety Board has developed the ALARP guidelines, an international standard widely used to determine whether risks are acceptable(Faber and Stewart, 2003; Zhou and Liu, 2012) .

The ALARP principle divides the risk into three areas: intolerable, ALARP, and broadly acceptable areas. The boundaries of the three areas are the upper line and bottom line of the risk. In this paper the optimization objective is determined according to its evacuation exposure risk for a specific original evacuation plan. If the risk belongs to the intolerable area, the original plan should be optimized to reduce the risk to the ALARP area. If the risk belongs to the ALARP area, the original plan should be to reasonably optimize if the cost range permits. If the risk belongs to the broadly acceptable areas, the risk can be ignored, and the original plan can be directly recommended to the decision makers.

The determination of the two risk thresholds in ALARP is industry-related. The probability of industrial accidents, accident consequences, and willingness factors should be considered (Jonkman et al., 2003; Lindell, 2000; Lindell et al., 2006). However, when we talk about "evacuation exposure risk", we take the occurrence of accidents as the premise to evaluate different evacuation plans in the same evacuation scenario. This makes the existing standards (for example, Jonkman et al., 2003) not applicable for determining the acceptable level of evacuation exposure risk. However, in order to illustrate the method proposed in this paper, the case study (see *Section 4*) works on the assumption that the decision makers give the thresholds.

3.6. Optimization of evacuation plans

In this section, an optimization method of the evacuation plans based on a heuristic concept is proposed. First, the EEPA is partitioned into several ESRs. Then the proposed evacuation exposure risk assessment method is used to evaluate the original evacuation plan according to the ALARP principle. Thus, the optimization objective of the original evacuation plan of each ESR is determined. Finally, the optimized evacuation plan is determined through the cost and benefit analysis of evacuation exposure risk reduction. The flowchart of the evacuation plan optimization method is shown in Fig. 5.

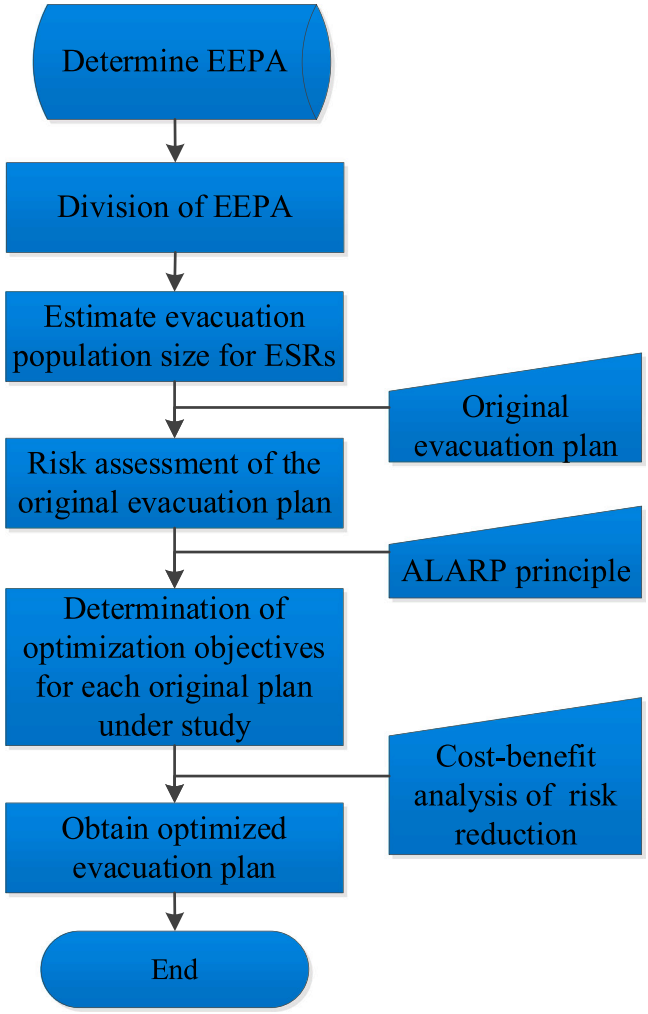


Fig. 5. Flow chart of the optimization of the evacuation plan.

4. Case study

4.1. Preliminaries

4.1.1. EEPA determination and division

The concept of EEPA in this paper is derived from the emergency planning zone. Therefore, deciding on the EEPA can refer to the existing determination method of the emergency planning zone, for example, the method mentioned in (GB/T 35622-2017, 2017) and Tawil et al.'s method for determining the actual evacuation area of radioactive substance release accidents in nuclear power plants (Tawil et al., 1987). The surrounding terrain and administrative boundaries should also be considered.

In this section, a chemical plant in Tianjin, China, is taken as an example. This paper mainly refers to the geographical location of the chemical plant, and extracts the chemical plant's two main characteristics: large storage of hazardous chemicals and dense of surrounding residential areas. And according to the results of statistical analysis on the causes of leakage of hazardous chemicals in China (Hou et al., 2021; Pan and Jiang, 2002), the toxic gas release scenario in a chemical plant is constructed. Suppose that a spill of liquid ammonia from a storage tank occurred at the position shown in Fig. 6. The release parameters are shown in Table 1. The liquid ammonia release model proposed by (Ding et al., 2007) can be used to estimate the concentration distribution of ammonia gas after the release (Fig. 7). According to this model, the dispersion of gaseous ammonia begins after the flash vaporization process of the liquid ammonia. During the initial period of gaseous ammonia dispersion, the ammonia concentration distribution is unstable. The Gaussian plume model and Gauss smoke cloud model can be used in the calculation. After a certain period, the concentration distribution of ammonia gas stabilizes. Therefore, the Gaussian plume model can be used in the calculation. The calculated stable concentration distribution of ammonia gas is shown in Fig. 6. It is assumed that the thresholds of broadly acceptable evacuation exposure risk and intolerable evacuation exposure risk given by the decision maker are 10^{-1} and 10^{-8}, respectively.

Based on the above method mentioned, the EEPA determined here is a combination of a circle area and a fan-shaped area according to the preliminary calculations of the toxic gas dispersion (Fig. 6). The location of the release source is taken as the central point. The radius of the circular area is set at 300 m due to the high level of toxicant concentrations within this radius. In addition, the downwind distance of toxic gas dispersion is approximately 5000 m. However, the coastline was already in contact with the boundary 2600 m downwind of the release source. Therefore, the radius of the fan-shaped area is set as 2600 m. The central angle is set as 90°, which refers to Lindell and Perry (1992) Chapter 8 (Protective action selection and implementation). The EEPA exits are set at the junction of the boundaries of the EEPA and the external roads. According to the EEPA partitioning method proposed in *Section 2.2* , the EEPA is divided into 26 ESRs. ESR 1, 2, 3, and 4 are industrial parks, and the occupants are primarily workers. Other ESRs are residential areas, and the occupants are primarily residents.

Table 1
Release parameters and meteorological conditions.

Release parameter		Meteorological parameters	
Volume (m^3)	50	Average temperature (℃)	25
Release shape	Circular	Average wind velocity (m/s)	3.5
Release hole diameter (mm)	40	Predominant wind direction	South wind
Height of release port (from the ground) (m)	2.4	Solar radiation intensity	Weak
Height of release port (from liquid level) (m)	1.3	Pasquill Stability Class	D
Internal pressure (pa)	1040,000		
Density (kg/m^3)	617		
Release duration (min)	65		
Release rate (kg/s)	3.23		
Measured height (m)	1.67		

4.1.2. Estimation of population to be evacuated and individual parameter settings

According to the survey data of personnel composition and per capita floor area of the Tianjin Binhai New Area (Hu, 2018), the size of the population to be evacuated in the EEPA is estimated to be approximately 6121 people, and the proportion of personnel types in each ESR are determined (Table 2). Based on empirical data(GB/T 10000–1988, 1988, GB/T 26158–2010, 2010; Shi et al., 2009; Gordon et al., 2010; Hurley et al., 2016), the parameters of the human body are set (Table 3). It should be noted that, due to the small proportion of older people among the research objects in this paper, the influence of gender was not

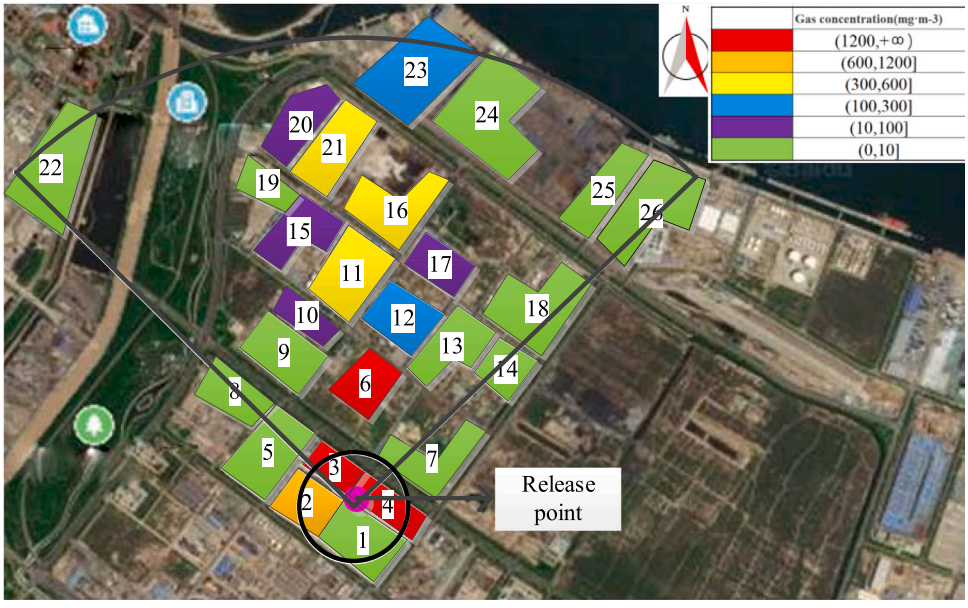


Fig. 6. Division of the EEPA and gas concentration distribution of each ESR.

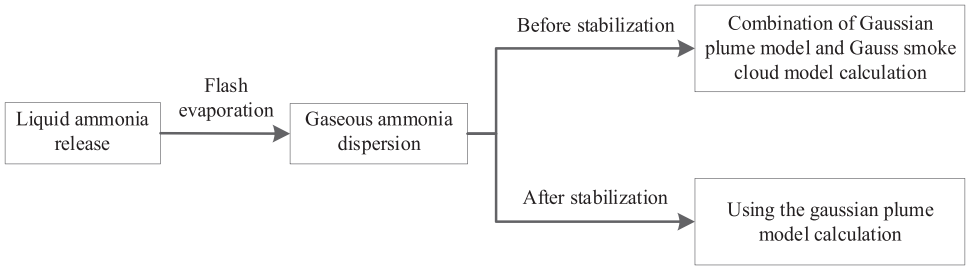


Fig. 7. Calculation method of concentration distribution of ammonia gas.

Table 2
Proportion of personnel types in each area.

	Personnel type	Percentage (%)
ESR 1.2.3.4	Man	70
	Woman	30
	Child	0
	Older	0
Other ESRs	Man	42.15
	Woman	40
	Child	12.07
	Older	5.78

Table 3
Parameter settings of personnel characteristics in each ESR.

Personnel type	Height (m)	Shoulder breadth (cm)	Walking speed (m/s)
Man	1.60–1.80	39.5–46.9	1.19–1.30
Woman	1.50–1.70	36.3–43.8	0.98–1.24
Child	1.30–1.60	31.9–40.2	0.49–1.08
Older	1.50–1.68	35.9–44.2	0.48–1.04

taken into account when setting the walking speed. The above parameters are then fed into the evacuation model proposed in *Section 2.2* to help simulate and analyze the evacuation of the population in the EEPA.

4.2. Evaluation results of original evacuation plan

In the original plan, the time between the release's occurrence and the evacuation order's issuance is set as 30 min. In order to test and analyze the process from the decision maker issuing the evacuation order to the people deciding to evacuate under the conditions that the social network of the crowd is low in clustering, the transmission network of the evacuation warnings is built based on the ER random network (Wang et al., 2012). It is assumed that people will decide to evacuate to the destination once the information value they receive exceeds the evacuation threshold l_{evi} (*Section 2.1*). In rapid onset disasters, people usually have preparation time to look for separated family members, gather emergency supplies, protect property, and help others in addition to seeking additional information and warning others (Lindell et al., 2015, 2019a, 2019b). However, as decision-makers, they hope that people can evacuate immediately in accordance with the evacuation order, and shorten the preparation time as much as possible. Therefore, the hypothesis of this paper is set in an ideal situation, that is, the public can evacuate immediately after deciding to evacuate. Based on the above assumptions, without considering the time lag and duration for each communication, the increasing trend of the percentage of people receiving evacuation warnings in the EEPA is obtained (Fig. 8). Pathfinder can be utilized to determine the evacuation path of each individual. In its algorithm, the individual evacuation paths are comprehensively determined based on the travel distance and the time before the individual reaches the exits (Tang, 2015). After calculating, the path assignment results of each ESR are shown in Fig. 9. According to the simulation results, the change trend in the percentage of people

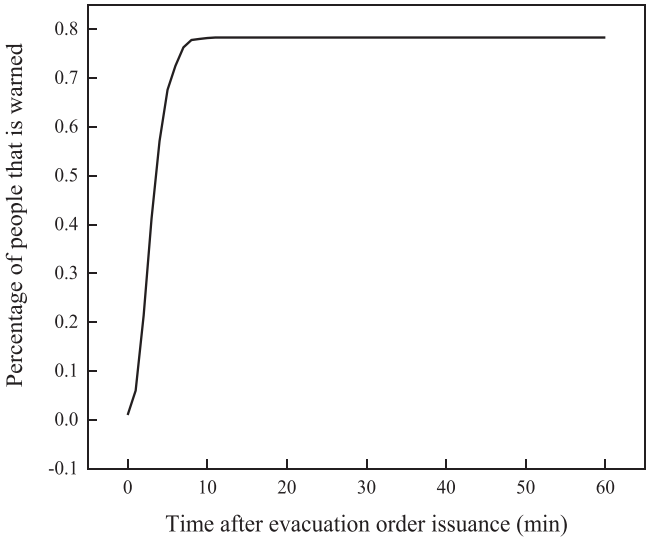


Fig. 8. Changes in the percentage of people warned within the EEPA.

evacuated and not evacuated from the EEPA can be obtained (Fig. 10). Fig. 10 shows that evacuation was completed after nearly 55 min. In the late evacuation stage, some people stayed still and failed to evacuate because they did not receive the evacuation warnings, or the received evacuation information value did not reach the evacuation threshold. Based on the calculation method of the evacuation exposure risk proposed in *Section 3.4* , the evacuation exposure risk partition of each ESR under the original evacuation plan is obtained (Fig. 11).

As shown in Fig. 11, for the ESRs located downwind of the spill, the closer they are to the release point, the higher the evacuation exposure risk is for them. For the ESRs with the same distance from the source, the closer they are to the axis of the fan-shaped area, the higher the evacuation exposure risk is for them. The ammonia concentrations and the evacuation exposure risk in each ESR are compared in Fig. 12. For most ESRs, the higher the ammonia concentration is, the higher the evacuation exposure risk. However, exceptions remain. For example, the ammonia concentrations in ESR 16 are higher than in ESR 21; however, the evacuation exposure risk of people in the former is lower than that in the latter. Although the differences in toxic gas concentrations and the evacuation exposure risk between ESR 16 and ESR 21 are minor, these minor differences may have deeper reasoning. It is found that ERS 21 is far away from the risk area exit, so the time that it takes people traveling within ESR 21 is longer than that within ESR 16. So, the overall dose of toxic gas inhaled by the personnel in ESR 21 is higher than that in ESR16. Therefore, the evacuation exposure risk of people in the former is higher than that of the latter. Thus, the evacuation exposure risk of ESRs is determined not only by the toxic gas concentration, but also by the duration of exposure.

According to the decision makers' thresholds of the evacuation exposure risk (See *Section 4.1*), the demand map of the evacuation

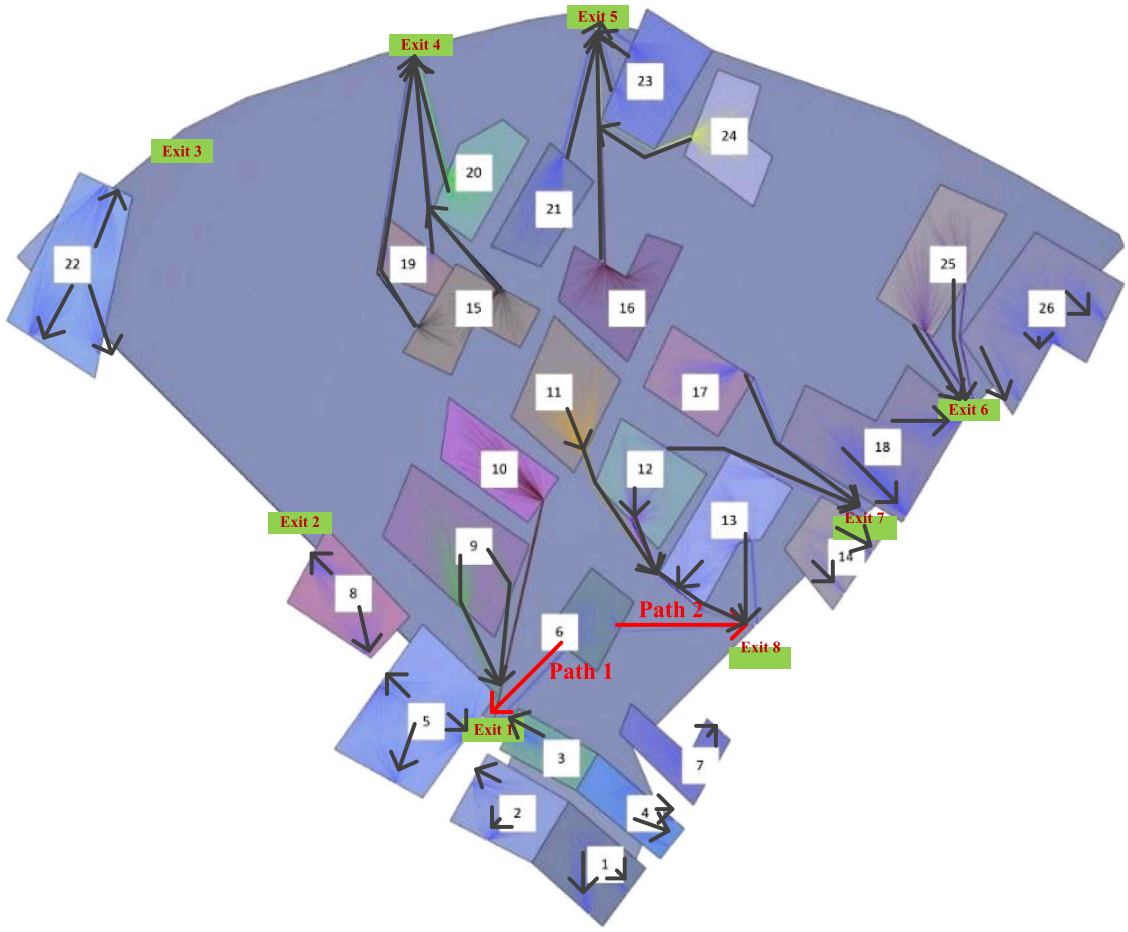


Fig. 9. Path assignment results of each ESR.

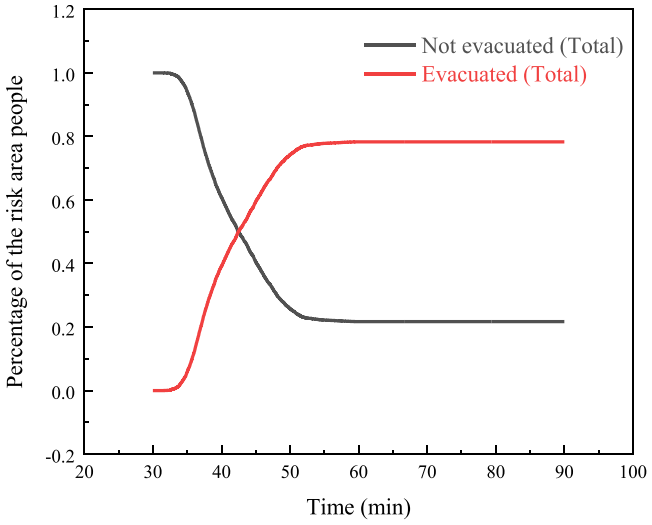


Fig. 10. Changes in the percentage of people that have evacuated and have not evacuated from the EEPA.

exposure risk reduction for the EEPA can be obtained (Fig. 13). For ESRs 2, 3, 4, and 6, the evacuation exposure risk of the population is higher than the ALARP's upper threshold (10^{-1}), and therefore corresponding measures should be taken to reduce the evacuation exposure risk to the ALARP area. For ESRs 11, 16, and 21, the evacuation exposure risk belongs to the ALARP area, reasonable measures should be taken to reduce the evacuation exposure risk within an acceptable financial cost range. For the other ESRs, their risk is lower than the lower threshold (10^{-8}), implying that the evacuation exposure risk of the ESRs under the original evacuation plan is acceptable, and thus additional evacuation exposure risk reduction measures are not required. Therefore, only the evacuation strategies for ESRs 2, 3, 4, 6, 11, 16, and 21 should be reformulated.

4.3. Analysis of influencing factors of evacuation exposure risk

According to the analysis in *Section 4.2* , the factors affecting the evacuation exposure risk of each ESR are not only the concentrations of toxic gas. Therefore, the quantitative assessment method of evacuation exposure risk is used to verify the rationality of some of the optimization strategies (optimization of the evacuation path selection, release time of the evacuation order, and the warning diffusion model). Therefore, the impact of the above optimization strategies on the evacuation exposure risk will be tested.

4.3.1. Influence of evacuation route selection on evacuation exposure risk

The evacuation route assignments of the original evacuation plan for each ESR are shown in Fig. 9. In the case of an evacuation, people in some ESRs face multiple evacuation paths. For example, there are two evacuation paths with different evacuation exposure risks in ESR 6. The evacuation exposure risk of route 1 is 0.55, and that of route 2 is 0.52. Hence, the selection of the evacuation path is also one of the factors affecting the evacuation exposure risk of ESRs.

According to the analysis in *Section 4.2* , the evacuation plans for ESRs 2, 3, 4, 5, 11, 16, and 21 should be reformulated. Furthermore,

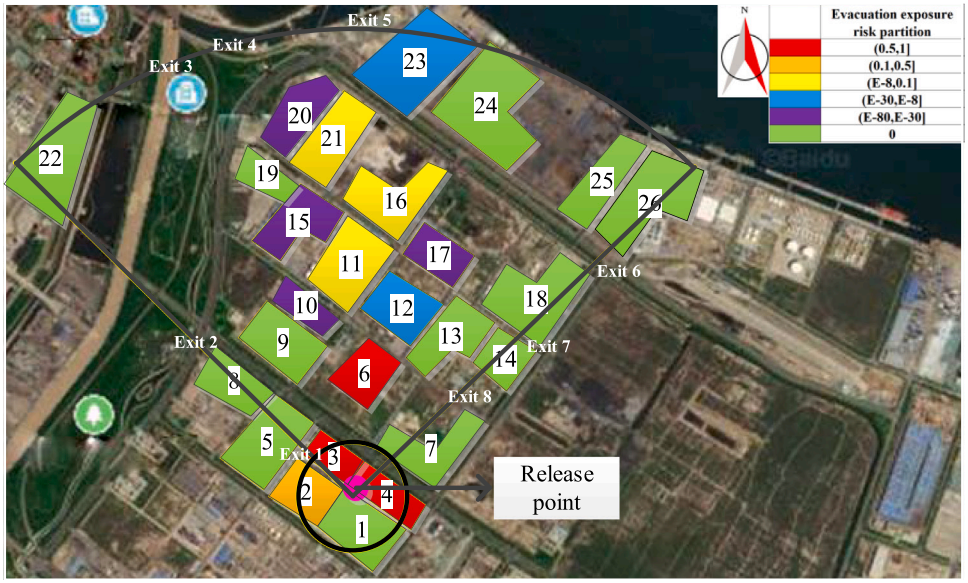


Fig. 11. Evacuation exposure risk partition of each ESR under the original evacuation plan.

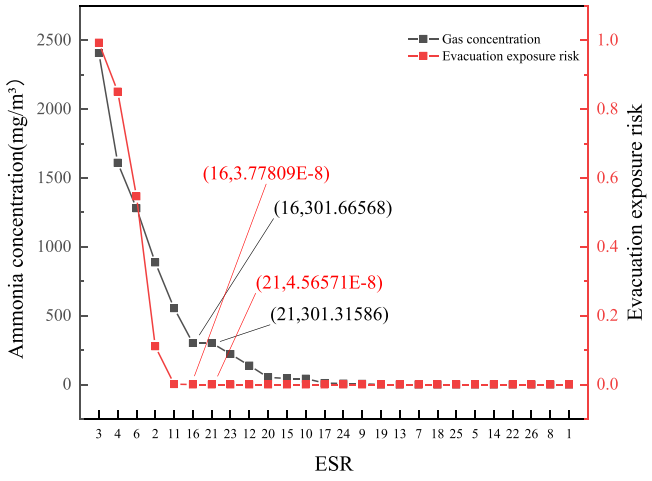


Fig. 12. Comparison of ammonia concentrations and the evacuation exposure risk in different ESRs. Note: The ESRs on the horizontal axis are numbered in terms of their ammonia concentrations in descending order.

people in ESRs 2 and 4 can directly reach the EEPA exits through the risk area exits of the two ESRs, and thus the evacuation path selection is not involved in the reformulation process of the evacuation plans. The evacuation exposure risk reduction tests are only conducted for ESRs 3, 6, 11, 16, and 21 (Fig. 14). The EEPA has a total of eight exits. During the test, there was no congestion at these exits or along all the evacuation routes. This shows that the replacement of exits for the tested ESRs did not make the additionally generated traffic demand exceed the link capacity. The evacuation exposure risk of the five selected ESRs is tested, and the results are compared with that under the original evacuation strategy (Fig. 15).

As shown in Fig. 15, by changing the destination exits for the five ESRs tested, the evacuation exposure risk of some ESRs decreased, whereas others increased. The increase of evacuation exposure risk implies that the paths allocated to the ESRs under the original evacuation plan are already the optimal paths, and the reallocation of the evacuation paths increases the risk.

Among the five tested ESRs, ESR 21 undergoes the most significant reduction in evacuation exposure risk when the originally assigned evacuation route changes. According to Eq. (10), the evacuation exposure risk of people in an ESR depends on the toxic gas exposure during the evacuation. The exposure dose consists of the dose exposed when people stay in the ESR or travel along the evacuation path. Therefore, optimizing the evacuation route selection can significantly reduce the exposure dose of the individuals within these evacuation routes; therefore, the evacuation exposure risk may be effectively reduced. The original exit of the evacuation destination for ESR 21 is exit 5. When the evacuation destination exit is changed from exit 5 to exit 7, the exposure duration on the path increases, but the concentration of toxic gas on the evacuation path decreases. The final result is that the exposure dose of people traveling through the evacuation path is reduced, thus reducing the evacuation exposure risk. Therefore, in the process of emergency evacuation route allocation for toxic gas release accidents, the selection of the evacuation destination should be determined based on the principle of proximity and the exposure dose of people traveling along the evacuation route.

4.3.2. *Impact of the evacuation order issuance time on evacuation exposure risk*

Nowadays, some chemical plants have been covered with many sensors. A Complete sensor network and its early warning function can basically issue the alarm immediately after the accident. However, the function of the sensor network is far from ideal in actual hazardous chemical release accidents. Firstly, although covering a complete sensor network in chemical plants is the development trend of the chemical industry, there are still many small and micro enterprises in developing countries that have not achieved comprehensive sensor coverage. Secondly, sensor equipment may also fail due to its own failure or artificial destruction. In addition, major dangerous chemical release accidents in chemical plants can easily affect the surrounding residential areas. And these populated areas are not necessarily covered by complete sensor networks. Thus, it is incompletely reliable to only rely only on the sensor network to realize the evacuation warning function when formulating an emergency evacuation plan. Therefore, this paper studies the impact of the evacuation order issuance time on evacuation exposure risk.

The time when the release occurs is set as the Zero point t_0, and the release time t_1 of the evacuation order is set to change within the range of [0–30] min. Then the change of evacuation exposure risk with the release time of the evacuation order can be obtained (Fig. 16). When the evacuation order release time is earlier than that in the original evacuation plan, the evacuation exposure risk of each ESR is reduced; however, the reduction degree varies. For ESRs with low original evacuation

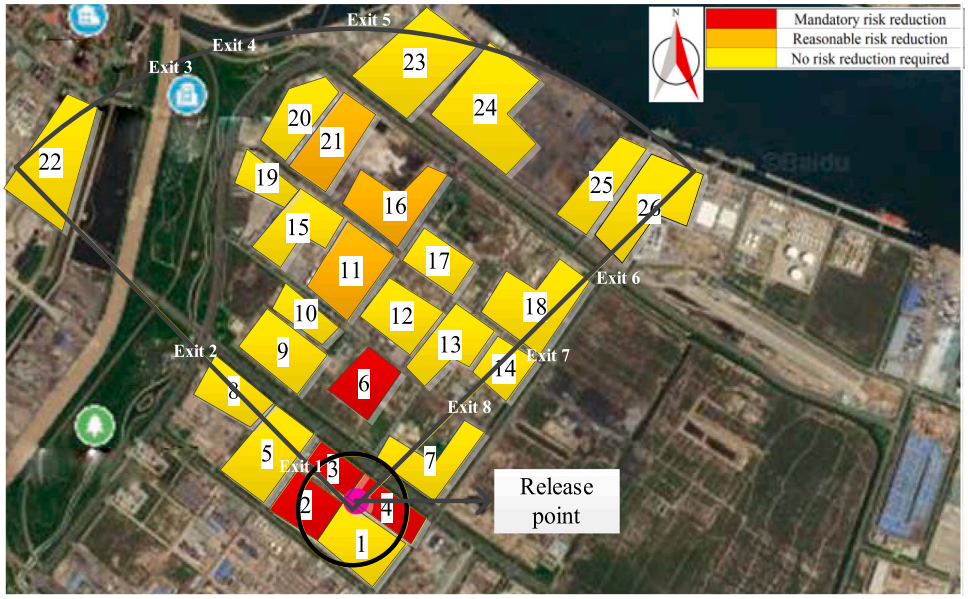


Fig. 13. Demand map of evacuation exposure risk reduction.



Fig. 14. ESRs to be tested for evacuation path optimization.

exposure risk (e.g., ESRs 16 and 21), if the evacuation order is issued 15 min earlier, the evacuation exposure risk is reduced to a relatively low level. For ESRs with high original evacuation exposure risk (e.g., ESR 2), if the evacuation order is issued 30 min earlier, the evacuation exposure risk remains at a higher level.

4.3.3. Impact of warning diffusion models on evacuation exposure risk

Two warning diffusion models are tested in this section. The ER random network represents the original warning diffusion model, and the WS network represents the optimized warning diffusion model. The optimization of the warning diffusion model could enhance the clustering degree of the population to be evacuated and enhance the transmission efficiency of the evacuation warnings (Wang et al., 2012). An obvious difference between the ER random network and the WS network is that the clustering coefficient of the latter is significantly higher than that of the former. The diffusion curves of the evacuation warnings corresponding to the above two types of networks can be calculated by Eqs. (3) and (4). The diffusion curves are input into Pathfinder software to analyze the increasing trend of evacuation flow (Fig. 17) and calculate the evacuation exposure risk before and after optimization of each of the ESRs as well as the degree of the evacuation exposure risk reduction after optimization (Fig. 18).

Fig. 17 compares the curves of the evacuation flow generation under the two warning diffusion models, namely, the ER random network and the WS network. The two curves reach their maximum in almost the same period. However, the final total number of people evacuated under the WS network is significantly higher than in the ER random network, thereby proving that the optimization of the warning diffusion model can improve the evacuation efficiency.

As shown in Fig. 18, the optimization of the warning diffusion model can reduce the evacuation exposure risk of the ESRs. The smaller the ambient ammonia concentration is, the higher the degree of risk

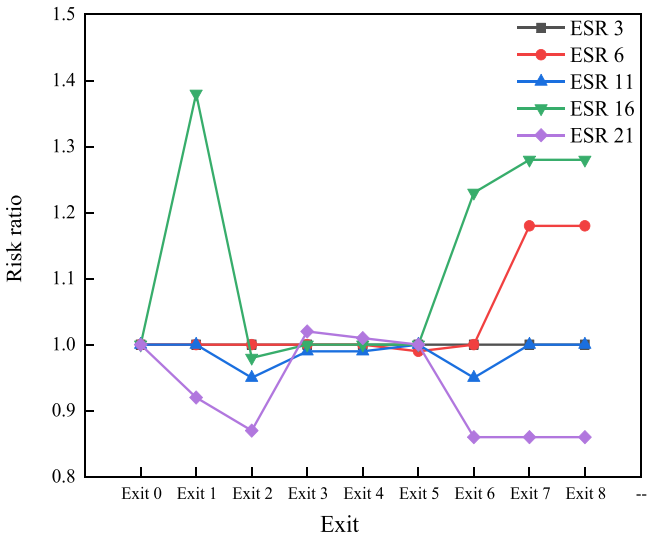


Fig. 15. Ratio of the current evacuation exposure risk and the original evacuation exposure risk after path selection optimization. Notes: Risk ratio is the ratio between the current and the original evacuation exposure risk; Exit 0 is the initial exit.

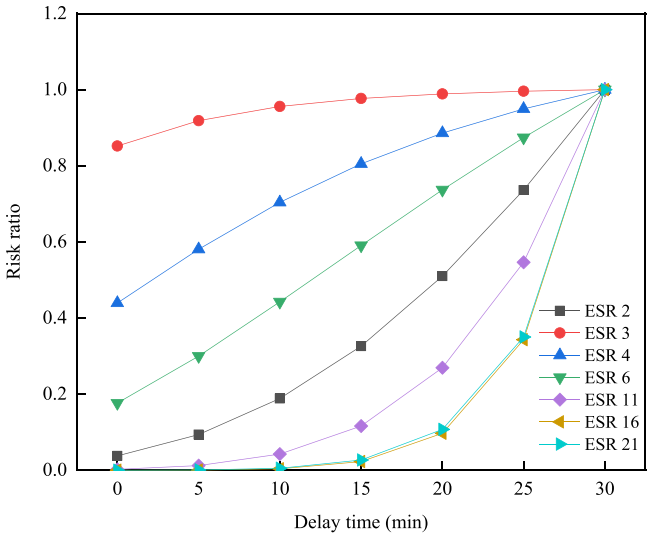


Fig. 16. Ratio of the current evacuation exposure risk and the original evacuation exposure risk in each ESR under the different evacuation order release times. Note: The risk ratio is the ratio between the current risk and the original evacuation exposure risk.

reduction becomes. From the above analysis of *Section 4.3.3* , it is known that the warning diffusion model optimization can reduce the time of personnel staying in each ESR, thereby reducing the evacuation exposure risk. However, under the premise of uniformly optimizing the warning diffusion model for all ESRs, the risk ratio reduction can change with the ammonia gas concentration in ESRs. Based on Eq. (10), it can be concluded that the time spent in the ESRs will increase the contribution to the evacuation exposure risk when its ammonia concentration decreases. Therefore, under the premise that the exposure duration is uniformly reduced in the ESRs, the lower the ammonia concentration is, the faster the risk ratio decreases. Based on this analysis, ESR 21 in Fig. 18 has the lowest ammonia gas concentration. Once the exposure duration in the ESR shortens, the decline in risk ratio is more apparent.

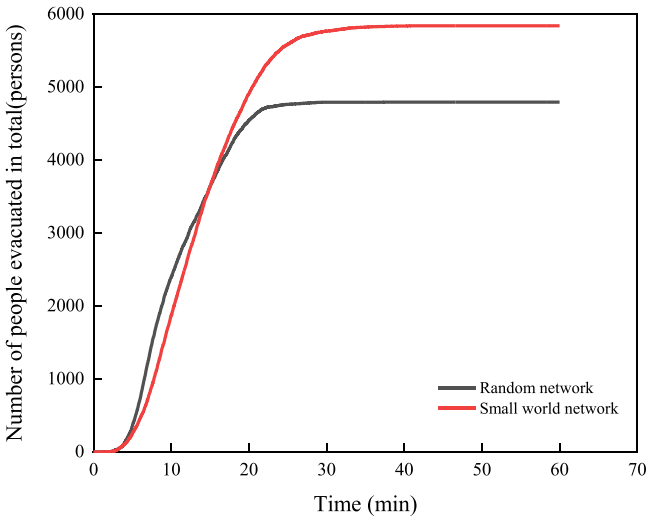


Fig. 17. Curve of evacuation flow generation under two warning diffusion models. Note: The time when the decision-makers deliver the evacuation order is set as the 0 min.

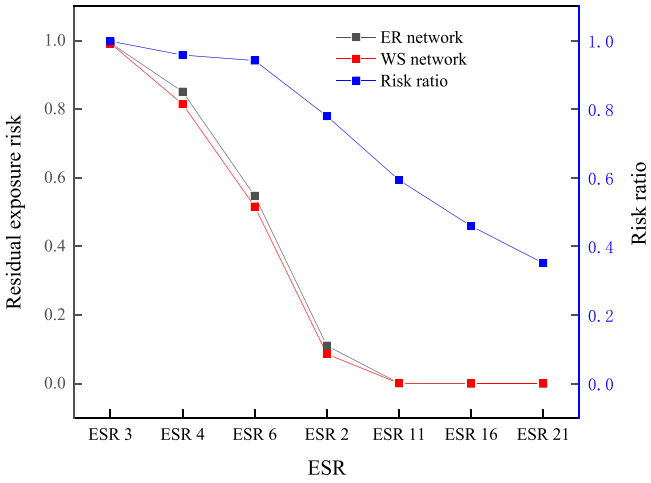


Fig. 18. Ratio of current evacuation exposure risk to the original evacuation exposure risk after optimizing the warning diffusion model. Notes: Risk ratio is the ratio between the current risk and the original evacuation exposure risk; The ESRs on the horizontal axis are numbered in descending order of their ammonia concentrations.

4.4. *Combination analysis of optimization strategies*

Three single optimization strategies (see *Section 4.3*) and combinatorial optimization strategies based on the single optimization strategies are compared and analyzed (Fig. 19). By comparing the optimization results of the three single optimization strategies, the optimization of the evacuation order issuance time has the most significant effect on reducing evacuation exposure risk. However, the single optimization strategies can only reduce the evacuation exposure risk of some ESRs to the target levels. The combined multiple optimization strategies can further reduce the evacuation exposure risk. According to the acceptable standards of the evacuation plans presented in *Section 3.5* and the risk thresholds given by the decision makers (see *Section 4.1*), the evacuation exposure risk of ESRs 2, 3, 4, and 6 should be reduced at least to the ALARP area ($[10^{-8}, 10^{-1}]$). However, Fig. 19 shows that only the risk reduction of ESRs 2 and 6 meets the optimization requirement of the evacuation plans. The evacuation exposure risk of ESRs 11, 16, as well as 21 should be reduced to the broadly acceptable area ($<10^{-8}$) if cost

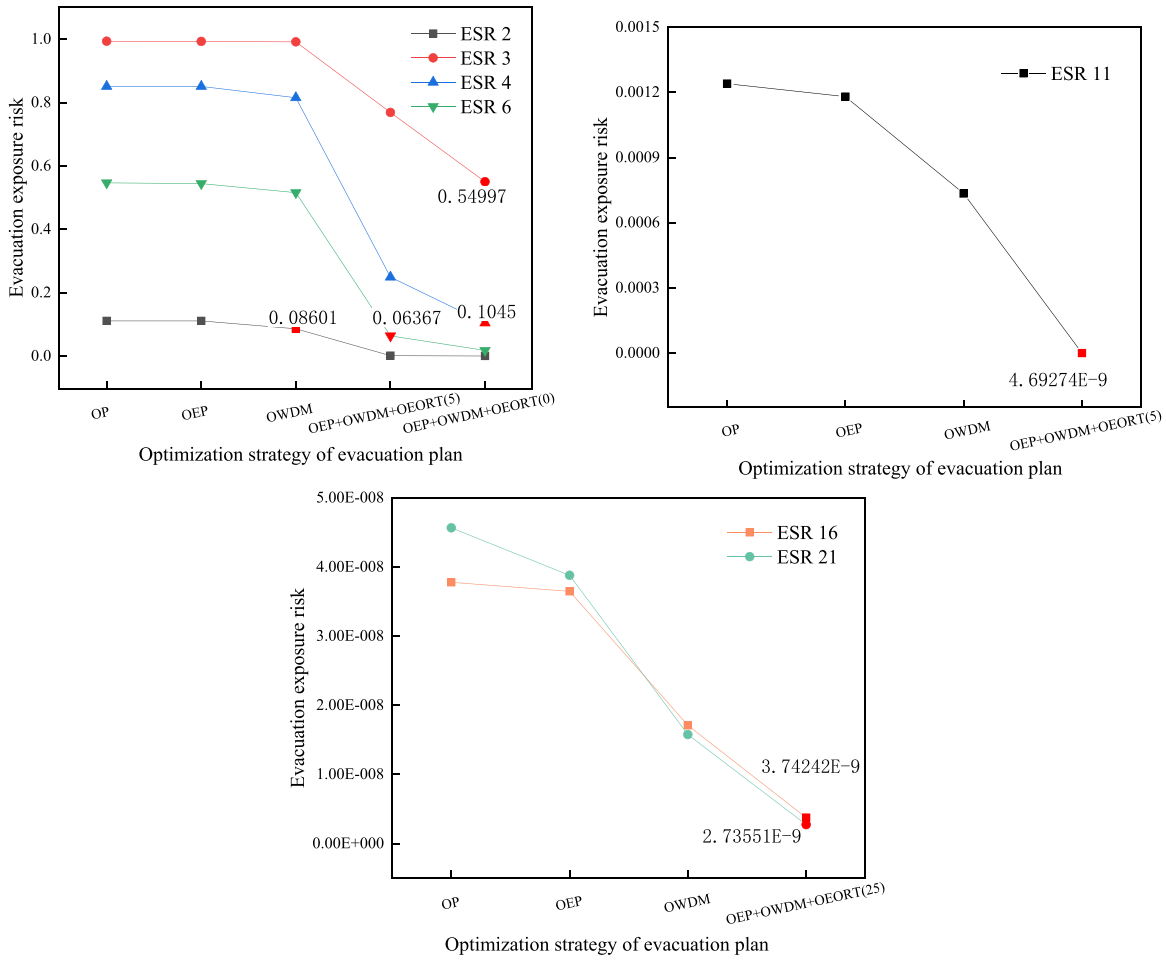


Fig. 19. Comparative analysis of the three single optimization strategies and the combined optimization strategies. Notes: OP represents the original plan; OEP is the optimization of evacuation path selection; OWDM is the optimization of the warning diffusion model; OEORT(x) denotes the optimization of the evacuation order release time (delay x minutes).

allows. Fig. 19 shows that the evacuation exposure risk reduction of all the above three ESRs meets the optimization requirements of evacuation plans.

Therefore, it is recommended to follow the following principles when selecting the optimization strategies of the evacuation plan. If several of the above optimization strategies (single or combined) can reduce the risk to the target level, it is recommended to choose a strategy that is easier to implement. If not, it is recommended to select the one that can minimize the evacuation exposure risk. According to this selection method, the optimization strategy recommendations for each ESR and the optimized evacuation exposure risk map can be obtained (Table 4 and Fig. 20). Fig. 20 shows that even if the three single or combined strategies proposed in *Section 4.3* are taken, the evacuation exposure risk level of people in ESRs 3 and 4 still cannot reach the risk reduction target of less than 10^{-1}. The remaining ESRs still require a combination of three single optimization strategies to achieve the risk reduction requirements set according to the ALARP principle.

If conditions permit, the evacuation order should be issued as soon as possible to reduce the evacuation exposure risk of all the ESRs. When early-warning resources are limited, the priority of the warning notification order for evacuating the population within the ESRs should be determined according to the evacuation exposure risk. Other auxiliary risk reduction measures should also be taken for ESRs with excessive evacuation exposure risk, apart from starting notifications as soon as possible. For the ESRs with low evacuation exposure risk, the evacuation order issuance time can be delayed within a reasonable range to optimize the allocation of early-warning resources.

Table 4
The optimal combination of the optimization strategies.

ESR	Optimization strategy	ESR	Optimization strategy
2	Optimization of warning diffusion model + issuing the evacuation order within 0 min	11	Optimization of warning diffusion model + issuing the evacuation order within 5 min
3	Optimization of warning diffusion model + optimization of evacuation path selection + issuing the evacuation order within 0 min	16	Optimization of warning diffusion model + optimization of evacuation path selection + issuing the evacuation order within 25 min
4	Optimization of warning diffusion model + issuing the evacuation order within 0 min	21	Optimization of warning diffusion model + issuing the evacuation order within 25 min
6	Optimization of warning diffusion model + optimization of evacuation path selection + issuing the evacuation order within 5 min		

According to the original evacuation plan and optimization strategies are shown in Table 4, each ESR's maximum allowed warning delay time can be obtained (Fig. 21). The evacuation order should be issued immediately after a release occurs for the ESRs in the downwind direction and closest to the release source (i.e., ESRs 2, 3, and 4). However, the evacuation order should be issued within 5 min for the ESRs in the downwind direction and far from the release source (i.e., ESRs 6 and 11). For ESRs 16 and 21, the evacuation order should be released within

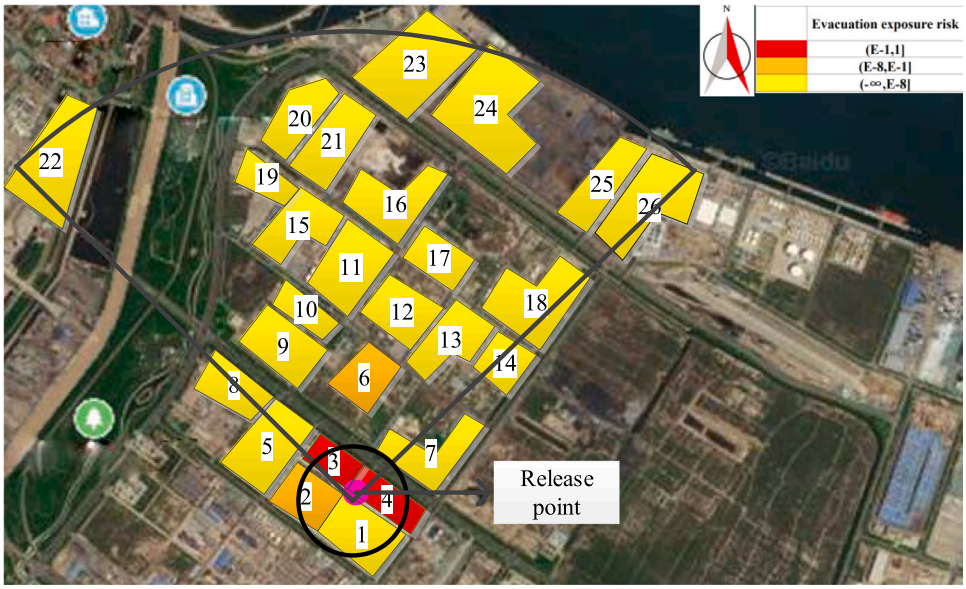


Fig. 20. Evacuation exposure risk of each ESR under different evacuation strategies.

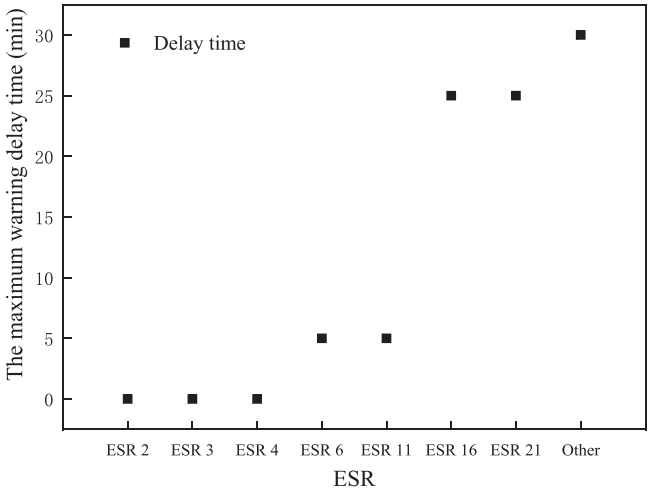


Fig. 21. Maximum allowed warning delay time for each ESR.

25 min. For the other remaining ESRs, the evacuation order should be delivered to the individuals within 30 min.

4.5. *Suggestions for optimization of evacuation plan*

According to the analysis in Sections 4.3 and 4.4, for the evacuation plans of some ESRs with evacuation exposure risk in the intolerable area and ALARP areas, their evacuation exposure risk can be reduced to the target level by optimizing the route selection, issuing early-warning order, and warning the diffusion model of the evacuation plan. In addition, we found that these ESRs are far from the release source or are not in the downwind direction. Meanwhile, for some ESRs with evacuation exposure risk in the intolerable area, the evacuation exposure risk cannot be reduced to the target level by optimizing the evacuation plans. These ESRs are generally close to the release source and are in the downwind direction. According to the risk assessment results, some auxiliary measures need to be added to reduce the evacuation exposure risk for those ESRs with high risk. For example, according to the policies and economic conditions of different regions, respiratory protective equipment can be provided to relevant people in order to reduce the dose of toxic gas inhaled by individuals during the evacuation. The personnel can be transferred after plant personnel have terminated the release or the toxic gas plume has dissipated; however, appropriate shielding conditions should be provided for sheltering in place. In addition, emergency rescue services can be provided to special groups (such as the disabled, the aged, etc.) during an emergency response to reduce the exposure duration of the individuals within the toxic gas.

For emergency response decision-makers, the above research gives a method for evacuation plan optimization. It tries to maximize the evacuation efficiency by making better use of available resources. When there are several evacuation routes to choose from, it can consider the evacuation distance and toxic gas concentration to provide a full evaluation of the evacuation exposure risk, which can assist people in choosing the least risky path. Furthermore, when evacuation resources are limited, the differentiated setting of the release time of the evacuation order and the warning diffusion model allows for the most efficient use of those resources.

It should be noted that the results obtained through simulation in this paper cannot be directly used as an emergency evacuation plan. However, as a low-probability and high-impact disaster event, there are few actual and usable cases of hazardous chemical release accidents. And the development of a detailed emergency evacuation plan is faced with insufficient accident data and experience. Especially in the study of the evacuation caused by the hazardous chemical releases, there are many people involved, and the evacuation range is wide. The evacuation drill alone is not enough to make up for the lack of relevant accident experience data. Therefore, this requires more refined and targeted risk analysis through the establishment of typical accident scenarios and computer simulation to provide a basis for an emergency evacuation plan. For example, as evacuation objects, due to the difference in work experience, chemical plant workers and the residents will have significantly different risk perception and evacuation behavior decisions. In the following research, the psychological characteristics of evacuees will be taken as modeling factors, and the influence of the psychological states of different evacuees on the formulation of emergency evacuation plans will be focused on.

5. Conclusions

This study uses a multiagent-based model to simulate the evacuation processes of a chemical plant in Tianjin, China. A risk assessment method for ambient evacuation is presented under the scenario of toxic gas release; therefore, an evacuation plan optimization process is

proposed. Without considering the time lag and duration of warning diffusion, and assuming that the public can evacuate immediately after deciding to evacuate, the evacuation scenario after the liquid ammonia release is simulated, and the effectiveness of the evacuation plan optimization strategies are verified. The conclusions are as follows:

(1) The concentration of toxic gas in the area and exposure duration are two parameters that determine evacuation exposure risk. Therefore, three factors should be considered while optimizing evacuation plans: evacuation route selection, evacuation order issuance time, and a warning diffusion model. The evacuation exposure risk may be effectively reduced when the optimal outcomes of the evacuation route selection can significantly reduce the exposure dose of an individual traveling through the evacuation path.
(2) The location of the release source and the direction of the toxic plume influences the effect of path optimization. In a specific accident scenario, some evacuation routes have inherent advantages to be far away from the diffusion of the toxic gas and do not need to be optimized, while some evacuation routes can be changed by restricting some roads and exits in order to reduce the exposure dose. Therefore, attempts to reduce the evacuation exposure risk of a given area by optimizing the evacuation routes are only suitable for specific areas.
(3) When the concentration of the toxic gas in the area drops, the contribution of exposure duration to the evacuation exposure risk increases. The exposure time of personnel in such areas can be shortened by issuing the evacuation order sooner and optimizing the warning diffusion model. Therefore, the two optimization strategies can effectively reduce the evacuation exposure risk of the areas with low toxic gas concentrations.
(4) The optimization strategies of the original evacuation plan for different evacuation areas are not the same. The strategies suitable for each area need to be determined according to the real situations of the evacuation areas. In general, if the area is far from the release source or is not in the downwind direction of the release source, one of the three single optimization strategies or combined optimization strategies can be adopted. For areas close to the release source and in the downwind direction, other auxiliary measures that can help reduce the evacuation exposure risk should be considered, such as providing individual protective equipment, building a shelter with excellent protective performance before an accident, or providing emergency rescue services.
(5) It is worth noting that for toxic gas release accidents, the smell of some chemicals, such as ammonia and hydrogen sulfide, provides an environmental cue that could cause people to evacuate even if they did not receive a warning. Such environmental cues would have the same effect as a broadcast warning directly to those in the risk area, thus producing earlier evacuation departure times in the same way as earlier official warnings. However, this paper didn't study the influence of environmental cue on emergency evacuation, which will be our next research direction.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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